

JAVA PLACEMENT PRACTICE QUESTIONS

56 Questions covering all Core Java topics from the Placement Syllabus
Conceptual • Theoretical • Coding • Output Prediction

Topics: Introduction • JVM • Program Lifecycle • IDE • Variables • Data Types • Operators • Type Casting • Input Handling • Access Modifiers • final • Object Class • if-else • switch • for / while / do-while • break & continue • Nested Loops

Tip: Answer without looking at notes first. For coding questions, write the code and dry-run it. For theory questions, practice explaining out loud as you would in an interview.

1. Introduction to Java

- Q1. Who developed Java and what was its original name?
- Q2. Explain the principle "Write Once, Run Anywhere" (WORA).
- Q3. List any 6 important features of Java and briefly explain each.
- Q4. Differentiate between Java SE, Java EE, and Java ME.

2. JVM Architecture

- Q5. What is the difference between JDK, JRE, and JVM?
- Q6. Explain the main components of JVM architecture (Class Loader, Runtime Data Areas, Execution Engine).
- Q7. Where are objects and local variables stored in the JVM memory model?
- Q8. What is the role of the JIT compiler?

3. Program Lifecycle

- Q9. Write the complete sequence of steps from writing a .java file to its execution.
- Q10. What happens during the "Linking" phase?
- Q11. Can you run a Java program without explicitly compiling it (from Java 11 onwards)? How?

4. IDE Setup

- Q12. Name two popular Java IDEs and one advantage of each.
- Q13. What is the purpose of setting JAVA_HOME?

5. Variables

- Q14. Differentiate between local, instance, and static variables with respect to: Scope, Lifetime, and Default values.
- Q15. Can a local variable be used without initialization? Why or why not?
- Q16. What is the output of this code?

```
public class Test {  
    static int x = 10;  
    int y = 20;  
    public static void main(String[] args) {  
        Test t1 = new Test();  
        Test t2 = new Test();  
        t1.x = 30;  
        t1.y = 40;  
        System.out.println(t2.x + " " + t2.y);  
    }  
}
```

6. Data Types

- Q17. List all 8 primitive data types with their sizes.
- Q18. What is the difference between int and Integer?

Q19. Explain Autoboxing and Unboxing with an example.

Q20. Why is char 2 bytes in Java while it is 1 byte in C?

7. Operators

Q21. What is the difference between / and % operators?

Q22. Explain the difference between && and & (short-circuit vs non-short-circuit).

Q23. Predict the output:

```
int a = 5;
System.out.println(a++ + ++a);
```

8. Type Casting

Q24. What is Widening casting? Give an example.

Q25. What is Narrowing casting? Why is an explicit cast required?

Q26. What will be the output of (byte)300? Explain.

9. Input Handling

Q27. Compare Scanner and BufferedReader. When would you prefer one over the other?

Q28. Write a small code snippet to read an integer and a full line of text using Scanner.

Q29. How do you read a line using BufferedReader?

10. Access Modifiers

Q30. Draw / explain the accessibility table for private, default, protected, and public.

Q31. Can a subclass in a different package access a protected member of the parent?

Q32. Why are instance variables often kept private?

11. final Keyword

Q33. What happens when you declare: (a) a final variable (b) a final method (c) a final class?

Q34. Can a final method be overloaded? Can it be overridden?

Q35. Why is the String class declared final?

12. Object Class

Q36. What is the default implementation of toString()?

Q37. Why should you override both equals() and hashCode() together?

Q38. What is the difference between == and .equals() for objects?

13. if-else

Q39. Write a nested if-else program to find the largest of three numbers.

Q40. Discuss when an if-else-if ladder becomes hard to read and how you would improve it.

14. switch

- Q41.** What are the limitations of the traditional switch statement (before Java 14)?
- Q42.** Write an enhanced switch expression that returns whether a day (1–7) is a weekday or weekend.
- Q43.** Does the arrow form (->) of switch allow fall-through?

15. for Loop

- Q44.** Write a program using nested for loops to print a right-angled triangle of stars.
- Q45.** What is the enhanced for-each loop and when can it be used?

16. while Loop

- Q46.** Why is the while loop called an entry-controlled loop?
- Q47.** Write a while loop that calculates the sum of digits of a number.

17. do-while Loop

- Q48.** Why is do-while called an exit-controlled loop?
- Q49.** Give a practical use-case where do-while is preferred over while.

18. break & continue

- Q50.** What is the difference between break and continue?
- Q51.** Can break be used outside a loop or switch? What happens if you try?

19. Nested Loops

- Q52.** Perform a dry-run of this nested loop and write the output:

```
for (int i = 1; i <= 3; i++) {  
    for (int j = 1; j <= i; j++) {  
        System.out.print(j + " ");  
    }  
    System.out.println();  
}
```

- Q53.** Write a nested loop program to print the multiplication table from 1 to 5.

Bonus Coding / Mixed Questions

- Q54.** Write a complete program that takes a number as input and prints whether it is prime (use any loop + break).
- Q55.** Explain the complete memory picture when this code runs:

```
public class Demo {  
    int x = 10;  
    public static void main(String[] args) {  
        Demo d = new Demo();  
        System.out.println(d.x);  
    }  
}
```

Q56. What will be the output and why?

```
Integer a = 100;  
Integer b = 100;  
Integer c = 200;  
Integer d = 200;  
System.out.println(a == b);  
System.out.println(c == d);
```

How to use: Attempt all questions without notes first. Write coding solutions on paper or in an IDE and dry-run them. Practice explaining theory answers out loud. After attempting, cross-check with the concept PDF.

All the best for your placements!